



- (iii) Make an aqueous solution of **FB 9** by adding a 5 cm depth of distilled water to a spatula measure of **FB 9** in a test-tube. Carry out the following tests on the aqueous solution of **FB 9** and record your observations in Table 3.2.

Table 3.2

<i>test</i>	<i>observations</i>
Test 1 To a 1 cm depth in a boiling tube, add aqueous sodium hydroxide, then ----- warm.	
Test 2 To a 1 cm depth in a test-tube, add a few drops of dilute hydrochloric acid, then add a few drops of aqueous chlorine. Empty and rinse the test-tube with water immediately after use.	

[2]

- (iv) Use your observations in (b)(i), (b)(ii) and Table 3.2 to deduce the formulae of the cations and anions in **FB 9**. If you are unable to identify an ion, write 'unknown'.

cations and

anions and

[2]

- (v) Write an ionic equation for the reaction in (b)(ii). Include state symbols.

..... [1]

[Total:14]

